Production note (1 page)

Deployment and scaling

- One uvicorn worker per container: models load once at startup (about 17 s locally, longer on CPU), then a warm-up request runs. Scale horizontally behind a load balancer; each replica needs its own copy of the models (about 2 GB for bge-m3 and the reranker, plus 2 x 0.5 GB classifiers).
- Inference runs in threads behind semaphores, and retrieval is serialized (thread safety of the shared encoder is UNVERIFIED). CPU retrieval took about 5.5 s in a single Docker sample versus about 0.3 s on the GPU host, so size replicas for the reranker, not the LLM.
- The LLM dominates cost and latency. Free-tier quotas (30 requests/min, 200K tokens/day per model) are a hard ceiling; production needs a paid tier or a self-hosted model.

Monitoring (from the JSONL trace, one line per request)

- p50/p95 per route and per span (preprocess, models, rag, sql, llm). Judge latency only on uncached LLM calls; the cache and pacing sleeps otherwise distort it.
- Route mix, refusal and clarify rates, and the clarify rate for natural questions (over-clarify was 7/46 in the eval).
- Invalid-citation and dropped-claim rates, SQL rejection rate by code, LLM 429 and timeout rates, error rate by `error_code`.
- Drift: route confidence histogram, share of queries below tau_route, retrieval gate probability.
- Alerts: error rate, p95 above target, LLM 429 spike, sudden growth in refusals.

Failure modes and responses

Failure	Behaviour today	Next step
LLM down or quota exhausted	`error/llm_unavailable`, retryable, no retry storm	Extractive fallback: return cited chunks without generation
Route model under-confident or wrong on natural questions (measured: 11 of 17 eval failures)	Clarify or refuse	Second signal: rescue `out_of_scope` with a customer-name/ID check for SQL and the retrieval gate for RAG; retrain on real traffic

Failure	Behaviour today	Next step
Claim verifier drops true claims (quotes of bullet-list chunks that differ from the text only in line breaks)	Refusal `no_evidence`	Find the normalization gap, make quote matching insensitive to line breaks and list markers; re-measure on a fresh set
Wrong-route answer (KB question answered with data)	Wrong answer, no signal	Cross-check SQL answers against route confidence; log for review
Poisoned or stale documents	Superseded docs dropped; retrieved text delimited as data; canary check	Ingest-time scan, document approval workflow
SQL timeout or schema drift	Refusal, or a loud startup failure for a missing DB	Schema check at startup; query timeout alerts
Index stale after a KB edit	Dense index rebuilt from a cache keyed on the chunks	Rebuild in CI and ship with the image

Security

Read-only SQL with allowlists and an authorizer, honeypot table, no LLM tools, secrets only in the environment, query text not logged by default (`TRACE_LOG_QUERY=1` opts in). Gaps: no authentication or per-user data scoping (any caller can ask about any customer), no rate limiting, no PII redaction in traces.

Cost

Per RAG request about 830 prompt and 160 completion tokens (one observed call); LLM price not computed (free tier). The classifiers cost CPU only. Cost control: the evidence gate and the clarify/refuse routes skip the LLM entirely (28 of the 70 eval cases finished in under 0.5 s in pass 1, i.e. without an LLM call), and responses are cached.

Updates and rollback

Version the models, KB, route data and prompts together (git tag plus an image tag). Roll back by redeploying the previous image; the DB is mounted read-only and untouched. Gate each model or KB update on the 70-case eval plus a fresh held-out sample; never tune on the eval set.